Measured Camera Datasheet

This datasheet reports the main camera of a Google Pixel 7a, measured from its raw DNG files.

Repository: `https://github.com/acrossman/ee513` (`project_1/`: code, `results.json`, and unit tests)

Vendor	Google	*Capture app*	Open Camera (Camera2 API), DNG
Model	Pixel 7a, main rear camera at 1x zoom	*CFA pattern*	RGGB
Data type	Raw DNG, 10 bit in `uint16`	*Black level*	64 DN (tag); 64.009 ± 0.001 DN measured
Sensor type	64 MP Quad Bayer CMOS, read out 2×2 binned (inferred)	*White level*	1023 DN (file and histogram)
Diagonal	1/1.73" (spec)	*Camera identity*	`lens_id` asserted on every series: `('Pixel 7a', 5.43, None, (4624, 3472))`
Pixel size	1.6 µm output (0.8 µm photodiode, spec)		
Lens	5.43 mm, f/1.9 (EXIF), OIS		
Resolution	4624×3472 pixels		

Manufacturer figures are from the Google Pixel tech specs (url: https://support.google.com/pixelphone/answer/7158570?hl=en). Because the raw file is binned, every per-pixel value below is per *output* pixel.

Operation points

OP	Measurement	Camera setting and data
OP1	Geometry	17 raw checkerboard views (9×6 inner corners, 19.2 mm squares)
OP2	Spatial resolution	slanted edge on a laptop screen, ISO 96–399: 9 captures at 1.10 m (center), 6 at 0.76–0.94 m (corners); the printed target (10 captures at 179 mm) for comparison; G1 plane (cycles per green sample)
OP3	Noise, dynamic range	ISO 45; 10 flat-field pairs, 1/50–3/10 s; 25 lens-capped frames, 1/40–16 s; G1 plane, central 128×128 patch
OP4	Color	ISO 399, incandescent light; 3 ColorChecker Classic Mini charts with a white sheet for shading

Results at a glance

Quantity	Measured	Manufacturer	Uncertainty from
Focal length (OP1)	3475.6 ± 5.9 px; 5.43 ± 0.005 mm	5.43 mm	OpenCV std. dev.
Diagonal field of view (OP1)	$79.5 \pm 0.1°$	80°	Propagated from f_x, f_y
Output pixel pitch (OP1)	1.56 ± 0.04 µm	1.6 µm	Bound covering the 2.4% gap to spec: about half focus breathing, the rest rounding
MTF50, center (OP2)	0.473 ± 0.017 cy/px (95% Nyquist)	–	Std. dev. over 36 edges (screen target)
MTF50, corner mean / worst (OP2)	0.22 ± 0.07 / 0.09 ± 0.09 cy/px	–	Bound: half the range between corners; top left defocused (without it 0.25 / 0.20 ± 0.04).
System gain K (OP3)	0.0702 ± 0.0011 DN/e^-	0.0788 (DNG profile)	Weighted fit covariance
Temporal dark noise σ_d (OP3)	$7.1 \pm 1.1\, e^-$	$7.19\, e^-$ (DNG profile)	Propagated from fit
Saturation capacity (OP3)	$13\,650 \pm 210\, e^-$	–	Propagated from K
Dynamic range (OP3)	10.8 ± 0.2 bit (65.1 dB)	–	Propagated from K and σ_d (SNR = 1 floor)
Dark current (OP3)	$0.957 \pm 0.016\, e^-$/s	–	Slope fit (G1 plane) and K
Mean ΔE_{00} after CCM (OP4)	3.77 ± 0.51 (worst 10.71 ± 0.40, red)	–	Std. dev. over 3 charts

Summary sheet for Operation Point: OP1 (Geometry, 17 raw checkerboard views)

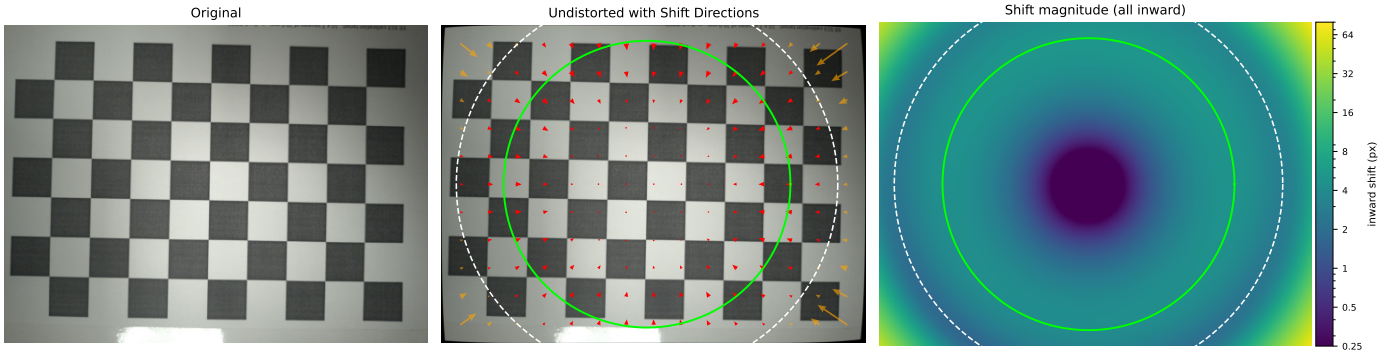


Figure. Left: original raw capture (`IMG_20261012_122925`). Center: the undistorted image with arrows (×15) from each grid point's original position to its undistorted position; orange arrows lie beyond the calibrated field. In the center and right images, the green circle is the boundary of field 0.54 where the k_1 and k_2 terms are equal; k_1 dominates inside it, k_2 (opposed by k_3) outside. The dashed circle is the boundary for field 0.72, the farthest calibration corner; beyond it the model is extrapolated and k_2 and k_3 nearly cancel. Right: size of the shift, inward everywhere: under 0.25 px at the center, 2–5 px across the calibrated field, up to 73 px at the frame corners.

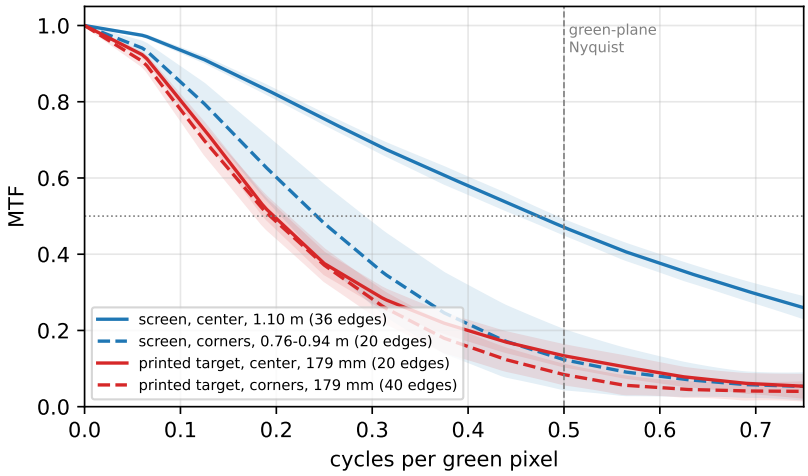
Method. `cv2.calibrateCameraExtended` on the raw captures (corners refined by `cornerSubPix`). The three terms, k_1, k_2, and k_3, are each tens of pixels and largely cancel: at field 0.72 they contribute +41, −72 and +34 px, a net 2.6 px.
Millimetres and pitch. The checkerboard square is 19.2 ± 0.1 mm (96 mm read on the 100 mm bar); this scales distances such as the board pose, not the intrinsics. The focal length in millimetres cannot be measured independently here, so the EXIF value is reported, and with it the pitch is $1000 \times 5.43/3475.6 = 1.562 \pm 0.003$ µm, 2.4% under the 1.6 µm binned spec. The reported ±0.04 µm covers that gap: about half is focus breathing (the calibration views were 191–981 mm away, which lengthens the effective focal length by about 1.3%), the rest is within rounding of the published values.
Against the manufacturer. The diagonal field of view is calculated from $2\arctan(5782.4/2/3475.6) = 79.5 \pm 0.1°$, 0.6% from the specified 80°.
Limitation, measured. No calibration corner lies beyond field 0.72, and past it the model's shift grows from a few pixels to 25 px at field 0.9 and 71 px at the frame corner (figure, right).

Performance

Focal length		
f_x	3477.0 ± 8.3	px
f_y	3474.2 ± 8.5	px
f (EXIF)	5.43 ± 0.005	mm
Principal point		
c_x	2238.3 ± 6.9	px
c_y	1736.2 ± 3.8	px
Radial distortion		
k_1	0.054 ± 0.005	
k_2	-0.27 ± 0.03	
k_3	0.35 ± 0.07	
Tangential distortion		
p_1	-0.0003 ± 0.0003	
p_2	0.0001 ± 0.0004	
Fit quality		
RMS reprojection	1.06 ± 0.30	px
views used	17 of 21	
Derived		
pixel pitch	1.56 ± 0.04	µm
diagonal FOV	79.5 ± 0.1	deg

Uncertainties: OpenCV `calibrateCameraExtended` standard deviations; RMS: spread of the per-view values (0.59–1.48 px).

Summary sheet for Operation Point: OP2 (Spatial resolution, slanted edge)



Performance

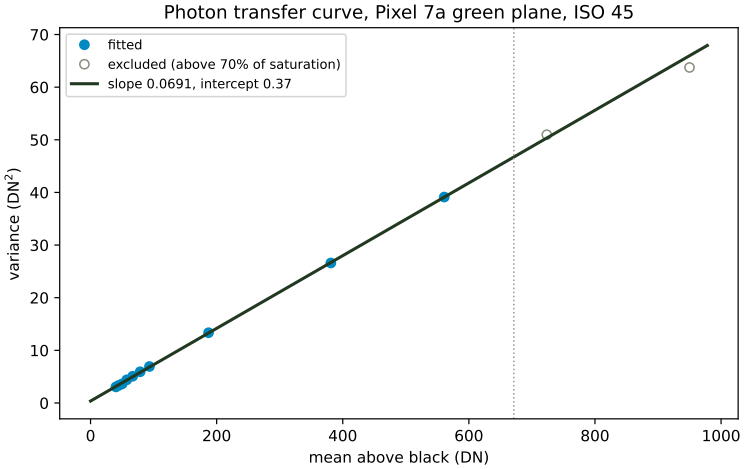
MTF50, screen (field; dist.)	**(cy/green px)**
center (0.16; 1.10 m)	0.473 ± 0.017
bottom right (0.77; 0.94 m)	0.25 ± 0.09
top right (0.80; 0.90 m)	0.27 ± 0.09
top left (defocused)	0.09 ± 0.09
bottom left (0.81; 0.76 m)	0.20 ± 0.09
corner mean	0.22 ± 0.07
center / Nyquist	95%
modulation at Nyquist	0.47 ± 0.02
Printed target at 179 mm	
center, field 0.31	0.189 ± 0.005
corner mean	0.182 ± 0.008
modulation at Nyquist	0.13 ± 0.03

Uncertainties: center, std. dev. over 36 edges (9 captures); corners, stated bound, half the range between them (Method); print, std. dev. over 10 captures.

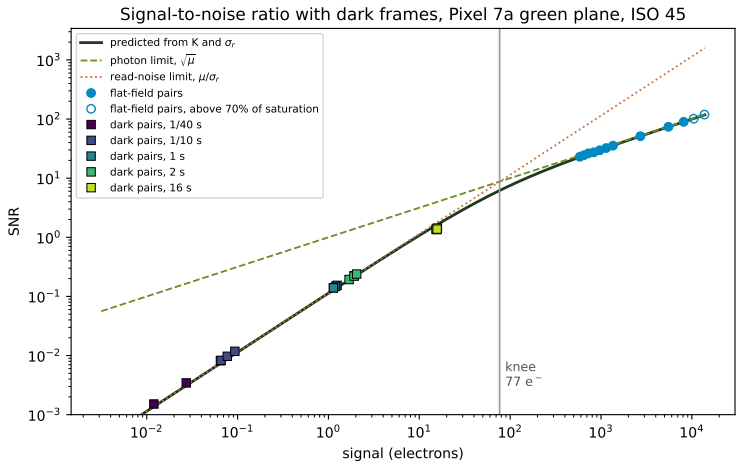
Mean MTF, band ±1 std. dev. over edges. Blue: the slanted edge on a laptop screen. Red: the printed target at 179 mm. Same ±8 px LSF (Line Spread Function) window for all.

Method. Slanted-edge MTF on the G1 plane, linear. The camera is measured on a square tilted 5° shown on a laptop screen. Originally this test was performed on a printed target but suspected toner spread was causing issues. The printed target performance values are provided as a comparison. The screen target was generated with the following constraints: 4:1 contrast, area anti-aliased edge (0.55 cycles per screen pixel, about 2 cycles per green pixel on the sensor, so negligible), no clipped pixels in any measured strip, distance from the square's known 95.9 mm size. Each side is measured from a strip of pixels that follows the edge, with no resampling; on synthetic edges that include the pixel aperture this reads −3.3% to +3.6% of the truth. **Center:** MTF50 is 95% of Nyquist, so the center is limited by sampling, as the diffraction cutoff (957 cycles/mm at f/1.9, three times the 320 cycles/mm Nyquist) predicts. **Corners:** The top-left pair was out of focus. Each corner is one setup (a pair of shots at 0.66–0.94 m), so each corner value is given a bound, half the range between the corners (±0.09 with all four). Omitting the top-left pair as a poor-quality capture changes the corner mean to 0.25 ± 0.04 (20 edges) and the worst corner to bottom left, 0.20 ± 0.04.

Summary sheet for Operation Point: OP3 (Noise, ISO 45, G1 plane)



Photon transfer curve. Half the pair-difference variance against the mean above black, 10 flat-field pairs on the central 128 × 128 G1 patch; open points (>70% of saturation) are excluded; slope K, intercept $K^2\sigma_d^2$.



Signal-to-noise ratio against signal, with the model $\mu/\sqrt{\mu+\sigma_d^2}$. Squares: lens-capped pairs (signal = dark current), in the read-noise regime below the knee that the flat fields never reach.

Performance

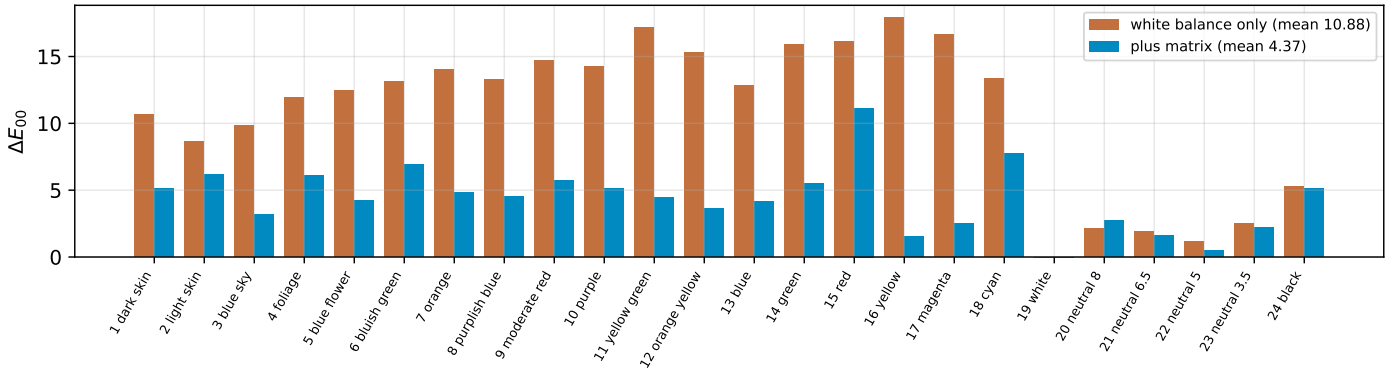
System gain and temporal dark noise		
K	0.0702 ± 0.0011	DN/e^-
σ_d ($\sigma_{y.dark}$, DN)	7.1 ± 1.1 (0.543)	e^-
Range; dark current		
SNR_{max}	116.8 ± 0.9 (41.4 dB)	
$\mu_{e.min}$	7.6 ± 1.1	e^-
$\mu_{e.sat}$	$13\,650 \pm 210$	e^-
DR	10.8 ± 0.2 (65.1 dB)	bit
knee σ_d^2	50 ± 16	e^-
$\mu_{I.mean}$	0.957 ± 0.016	e^-/s
Vendor DNG `NoiseProfile`		
K (σ_d, e^-)	0.0788, +12% (7.19)	DN/e^-

Method. Each point is the mean of a flat-field pair above the measured black level and half the variance of their difference, which cancels fixed pattern; the central patch avoids lens shading. The weighted fit is reported: on 200 simulated sensors, ordinary least squares, which treats every point as equally noisy, understates the gain uncertainty 3.6 times (here it gives $K = 0.0691 \pm 0.0002$, $\sigma_d = 8.75 \pm 0.64\,e^-$). The covariance is inflated by $\chi^2/\mathrm{dof} = 4.7$. Confirmation: repeated on 25 patches around the center, K reads up to 3% high off center; removing the light changes between the two frames of each pair (row bands and gradients) brings all 25 to 0.0699 ± 0.0005, the central value, and lowers χ^2/dof to 2.7, so the excess scatter is illumination, not noise. **Dynamic range:** EMVA 1288's criterion, $\mu_{e.\mathrm{sat}}/\mu_{e.\mathrm{min}}$ with $\mu_{e.\mathrm{min}} = \sqrt{\sigma_d^2 + 1/4} + 1/2$ (SNR = 1); a floor at σ_d gives 10.9 bit.

Why not a single frame. Single-frame variances give $K = 0.1178$ (71% high): PRNU and the illumination gradient grow with the signal, so their variance grows as signal squared. **Dark current:** slope of the G1-plane mean of 25 capped frames (about $15\,e^-$ at 16 s; temperature not recorded).

Vendor and independent code. The `NoiseProfile` read variance matches (0.320 vs. $0.248 \pm 0.071\,\mathrm{DN}^2$); its gain, 12% higher, does not (unexplained). `emva1288` v1.0.2: $K = 0.0693$, $\sigma_d = 6.6\,e^-$, 10.7 bit.

Summary sheet for Operation Point: OP4 (Color, incandescent, ISO 399)



Per-patch color error (CIEDE2000) before and after the 3×3 matrix, incandescent light, chart 1 of 3; the means in the legend are this chart's.

Method. Quarter-resolution RGB without demosaicing, black subtracted, divided by a white sheet; patch means over the middle half; gains from gray patch 21; 3×3 least-squares matrix to XYZ. Reference: "ColorChecker24 – After November 2014" (`colour-science`), the current formulation of the chart; the charts' date is unknown, so three were compared, and their matrices agree within 2.65°. The residual cannot reach zero: the camera's filters are not a linear combination of the eye's color matching functions.

Performance

ΔE_{00}, white balance only	
mean	10.64 ± 0.24
worst (yellow)	18.05 ± 0.12
ΔE_{00}, 3×3 CCM	
mean	3.77 ± 0.51
worst (red)	10.71 ± 0.40
Same CCM under other light (mean)	
LED	4.23 ± 0.22
outdoor	8.17 ± 0.47

Uncertainties: std. dev. over 3 charts.

Decisions and Verification

Decisions (and what each rejected)

- **Calibrate from raw DNGs, not JPEGs.** A phone's JPEG pipeline can apply its own distortion correction, so a JPEG calibration would measure that correction rather than the lens.
- **Measure on one green plane, linear, with the measured black level** (64.009 ± 0.001 DN; 0.01 DN from the tag, under $0.001\,\mathrm{DN}^2$ on the PTC (Photon Transfer Curve) intercept). Electrons are defined by this plane's gain, so dark current is fitted on it too; a fit over all four color channels had given 0.736 instead of $0.957\,e^-/\mathrm{s}$.
- **Measure the camera's MTF on a screen, not a print.** The printed target, 179 mm away, measured print and close range together (0.189); a slanted edge on a laptop screen at 1.1 m has no toner and gives 0.473. The print is kept as a comparison.
- **Report the weighted photon transfer fit** (ordinary least squares understates the gain uncertainty 3.6 times) and **EMVA 1288's dynamic range** (SNR = 1), stating the read-noise floor (10.9 bit) too.
- **Reject the first, hand-held flat-field series** and reshoot it with the phone immobilized and the paper taped down.
- **Use stated bounds where repeat captures cannot give the uncertainty:** the pixel pitch (± 0.04 µm, covering the 2.4% gap to the 1.6 µm spec, about half of it focus breathing at 191–981 mm) and each screen corner (half the range between corners, as each is one setup); the out-of-focus top-left corner is reported, with the values without it.
- **Scale the board from its printed 100 mm bar** (read as 96 mm, so 19.2 mm squares, not 20). **Keep nine frame-filling frontal views out of the calibration** (all parallel, and focused at 181 mm); they serve instead as an independent straightness test.

Verification

- **Known-answer tests, thresholds set first.** On 200 simulated sensors the gain scatters 0.7% and the read noise 24% (bias -5.3 ± 1.7%), so the pass criteria were 2% on gain and 5% on MTF50. The photon transfer code recovered a simulated sensor ($0.4266\,\mathrm{DN}/e^-$, $2.8\,e^-$) to -0.7% and $+5.7$%; the slanted-edge code, Gaussian edges to -1.7% to $+3.8$%; the screen's strip method, MTF50 0.18–0.41 to within 5% at 5–15° tilt. 65 unit tests on synthetic data: 62 pass, and 3 record known limits on purpose (the understated least-squares uncertainty; the strip method 6–8% low near 20° tilt; aliasing of short, shallow strips).
- **Independent code.** The `emva1288` package (v1.0.2) on the same DNGs picks the same 10 pairs and agrees on K to 0.3% (unweighted) and 1.3% (weighted) and on dynamic range to 0.1 bit; its lower read noise ($6.6\,e^-$) is its subtraction of quantization noise.
- **Failures found and fixed.** (a) The first flat-field series gave $K = 0.0764$ and a negative intercept; a bootstrap put the dark-frame value outside 2000 of 2000 resamples, so it was not scatter. Removing row bands and gradients from each pair's difference gives $K = 0.0685$ and $+0.357\,\mathrm{DN}^2$, and the same correction moves the reshot series by only 0.2%. (b) MTF50 was the lowest among comparable measurements; quick tests a reviewer suggested ruled out shake (horizontal edges 3.6 ± 0.8% softer) and lens falloff, and the cause was the 179 mm distance: board MTF50 rises from 0.16 at 191 mm to 0.42–0.45 at 500–980 mm, and a second print at 181 mm gives 0.203. (c) An earlier version of the MTF code had been validated (it read 10.5% low), not the one used. (d) The all-channel dark-current fit explained why two estimates of the 16 s dark signal disagreed (12 vs. $15\,e^-$). (e) f_x and f_y in millimetres had been converted with two different pitches (5.04 vs. 5.40 mm); with one, they are 5.432 and 5.427 mm. (f) The corner falloff was first called insignificant using a per-capture standard deviation; the standard error gives 3.8 ± 1.7%, $t = 2.3$. (g) The first undistortion example was a byte-identical copy of a calibration view. (h) A unit test found the handwriting filter taking a high-contrast mark for the edge; marks had lowered one corner's MTF50 from 0.159 to 0.121.
- **Model limits.** Beyond the last calibration corner (field 0.72) the distortion model over-corrects: a ruler bows 3.5 px before undistortion and 7.4 px the other way after, and the frontal grid lines straighten up to field 0.8 (RMS 2.45 to 2.00 px) but not beyond.
- **Cross-checks.** PTC intercept vs. dark-frame variance, 0.7σ; SNR vs. model, 0.9% median; field of view vs. spec, 0.6%; three color charts within $\pm 0.5\,\Delta E_{00}$; K on 25 patches, 0.0699 ± 0.0005 with illumination changes removed; `lens_id` and ISO asserted on every series.

Still unsure about

- The 12% (8σ) gap to the vendor's gain: illumination changes raise K off center, but not by this much at the center.
- Whether the 179 mm loss is print blur or close-focus lens performance; how steep the corner falloff is (one setup per corner, at mixed distances); distortion beyond field 0.72.
- The 16 s dark pairs sit about 15% below the predicted SNR: hot pixels, or dark current rising as the phone warmed (temperature was not recorded).

Use of AI tools. Claude (Anthropic) was used for organizing LaTeX, writing Python scripts for data visualization and unit testing, and as a technical writer. I reviewed the code against those reference examples, and when I could not follow a function's calls and parameters I simplified it until I could. The known-answer tests and cross-checks above are how I decided whether to trust the output. They are designed with synthetic data as a golden reference for the code. Much of my notes are processed by AI tools into more concise forms. I review and edit for clarity, but I don't rewrite every word so at times my voice is removed.

Resume line. Built an open-source Python camera characterization toolkit (lens calibration, slanted-edge MTF, photon-transfer noise model, color correction) and published a measured datasheet, with uncertainties, for the Google Pixel 7a main camera.